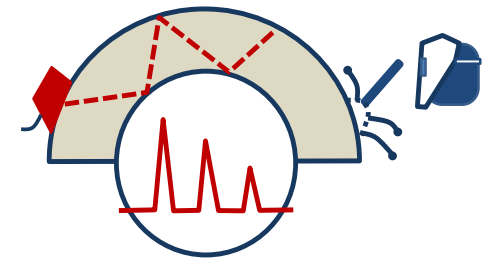


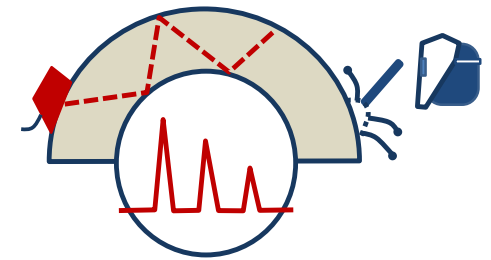
CWI — PART B

FINAL REVISION



المركز العربي لاستشارات
المواد الهندسية والتفتيش واللحام

STRUCTURAL STEEL CRITERIA

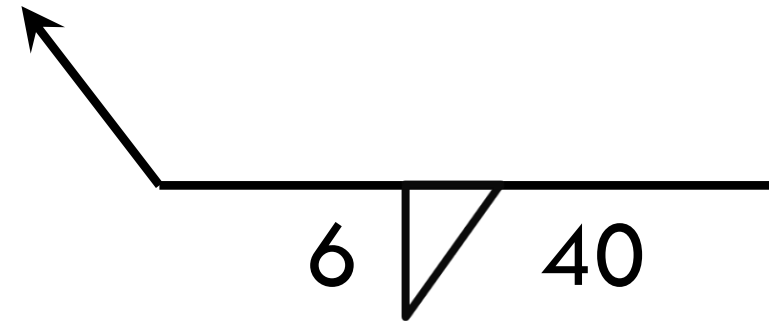


المركز العربي لاستشارات
المواد الهندسية والتفتيش واللحام

(1)

- Examine the lap joint between corners B and C and the welding symbol. Which one is true regarding the quality of this joint?
- (A) unaccepted – undersize
- (B) unaccepted – UC
- (C) unaccepted – weld length
- (D) unaccepted – wrapping around corner
- (E) none of the above

cracks

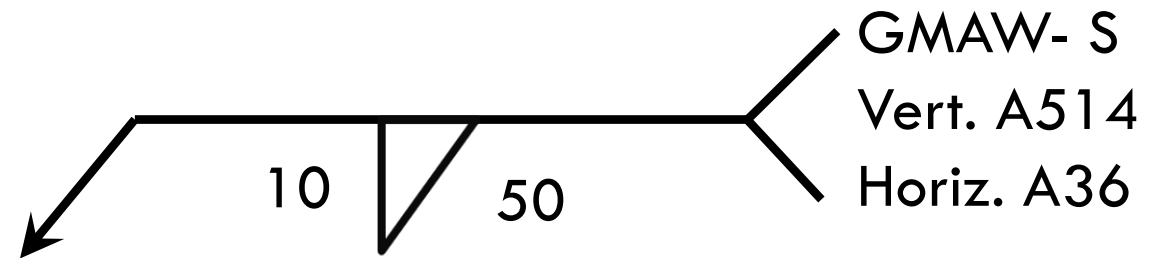


(2)

New

Structural Steel

- Examine the T-joint EF6 and the welding symbol. Acceptance criteria for this joint shall be based on visual inspection performed:



- (A) 3 days after of completion of the weld
- (B) 1 day after of completion of the weld
- (C) immediately after of completion of the weld
- (D) none of the above
- (E) time of inspection is not critical for structural steel

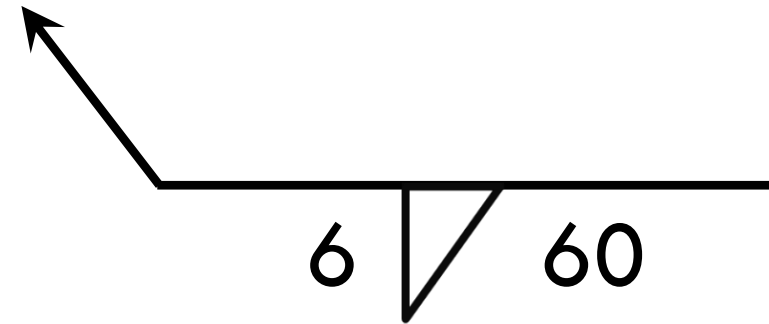
(3)

New

Structural Steel

■ Examine the lap joint EF-1 and the welding symbol. Which one is true regarding the quality of this joint?

- (A) unaccepted – undersize
- (B) unaccepted – UC
- (C) unaccepted – weld length
- (D) unaccepted – wrapping around corner
- (E) accepted

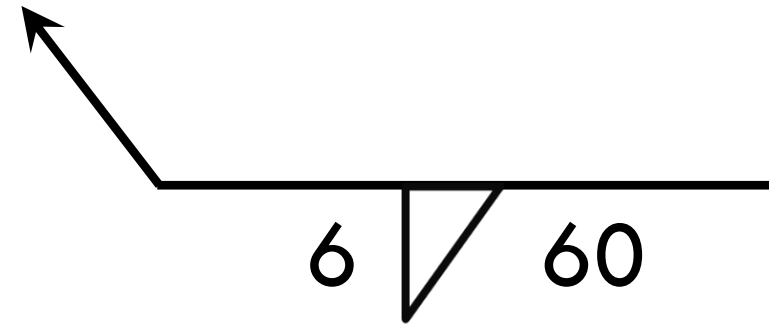


(4)

New

Structural Steel

- Examine the lap joint EF-3 at corner and the welding symbol. Which one is true regarding this joint?
- (A) unaccepted – undersize
- (B) unaccepted – UC
- (C) unaccepted – weld length
- (D) unaccepted – wrapping around corner
- (E) accepted



(5)

- Examine bend B1 for structural steel criteria, disregard any preparation or bending requirements, which one is true regarding B1?
- (A) accepted
- (B) reject – discontinuity > 3 mm
- (C) reject – corner crack results from porosity, > 3 mm
- (D) none of the above
- (E) B and C above

(6)

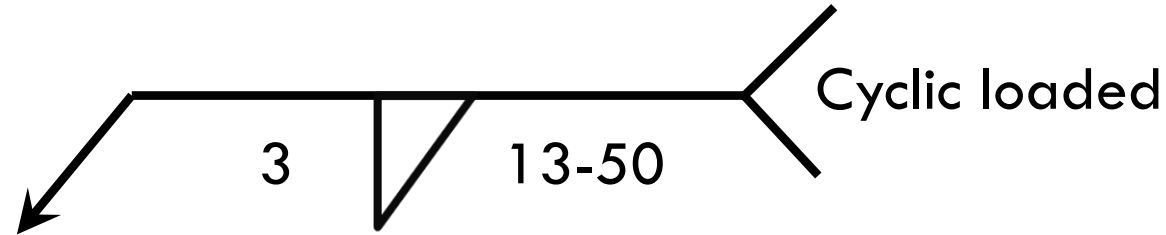
Examine EE-3 thermal cut edge roughness of steel structure for mechanized FCAW. Which one best describes it? ⁽³²⁾

- (A) Accepted
- ☒ (B) Unaccepted without additional preparation
- (C) Unaccepted because a machined edge is required
- (D) Unaccepted because of mill induced discontinuity
- (E) both B and D

(7)



- Examine lap joints at corners B and C according to the welding symbol shown:



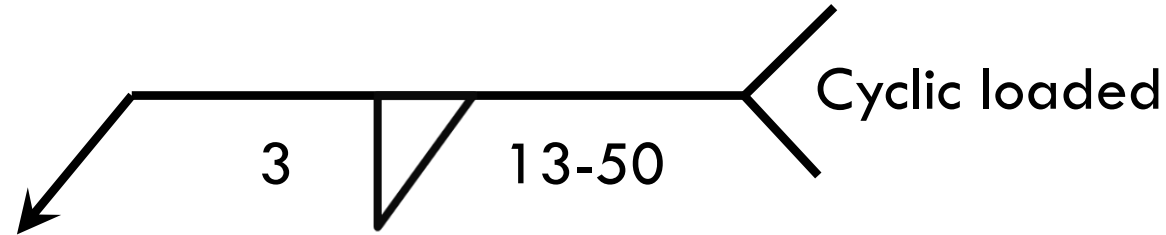
- Which one is true regarding starts and terminations at B and C?
- (A) unaccepted – no wrap around at C
- (B) accepted – around B
- (C) unaccepted – weld size
- (D) unaccepted – no hold back at corner C
- (E) both (A) and (B)

(8)

New

Structural Steel

- Examine lap joints at corners B and C according to the welding symbol shown:



- Which one is true regarding starts and terminations at B and C?
- (A) unaccepted – wrapping around corners
- (B) unaccepted – weld spacing
- (C) unaccepted – weld size
- (D) unaccepted – no hold back at corner C
- (E) both (A) and (B)

(9)

New

Structural Steel

- Examine bend B3 for structural steel criteria, disregard any preparation or bending requirements, which one is true regarding B3?
- (A) accepted
- (B) reject – discontinuity > 3 mm
- (C) reject – sum of discontinuities > 10 mm
- (D) none of the above
- (E) B and C above

(10)

New

Structural Steel

Examine T joints from EF1 to EF6. Which one has visible cracks? (2)

(A) EF2

☒ (B) EF3

(C) EF4

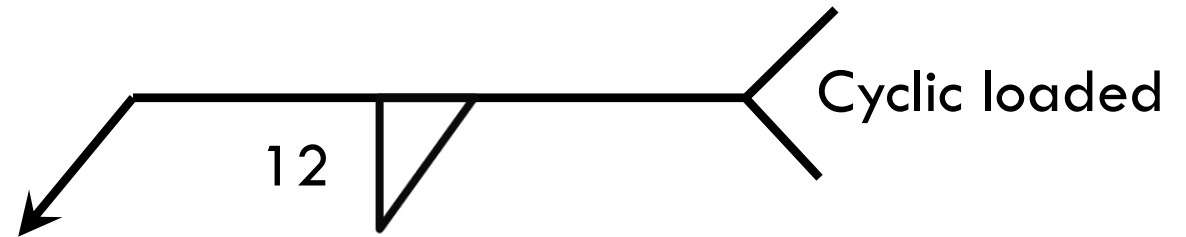
(D) EF5

(E) EF6

(11)



- Examine EF6 where its welding symbol on the DWG shown. Which one best describes it:



- (A) accepted
- (B) reject – due to undersize, undercut
- (C) reject – due to undersize, overlap
- (D) reject – due to undersize only
- (E) reject – due to overlap only

(12)

New

Structural Steel

- Examine the 6 mm partial joint penetration weld EG2-1. Is this joint accepted:
 - (A) reject – reinforcement > 2 mm
 - (B) reject – reinforcement > 3 mm
 - (C) reject – reinforcement > 5 mm
 - (D) accept – reinforcement ≤ 3 mm
 - (E) accept – reinforcement ≤ 5 mm

(13)

Examine EF2 T joint of steel structure with static loading: (7)

- (A) accepted
- (B) rejected – overlap
- (C) rejected – porosity
- ☒ (D) rejected – UC
- (E) rejected – unequal leg size

(14)

New

Structural Steel

The lap weld joint in EF-6 (specified size 3 mm) was made in a statically loaded structure. Is this acceptable joint? ⁽²⁶⁾

- (A) No, for weld size
- (B) No, for the weld termination at corner
- (C) No, UC
- ☒ (D) B and C
- (E) A and C

(15)

A structural steel weld EG1-5 was found to have excessive reinforcement and was repaired by grinding to remove excess metal. Is this repair acceptable? ⁽²⁷⁾

- (A) Yes, since excess metal is removed
- (B) Yes, since repair is done
- ☒ (C) No, since base metal was nicked
- (D) No, since middle point was not repaired
- (E) None of the above

The weld EG1-2 was made parallel to direction of tensile stress in statically loaded connections. Is this weld is acceptable? ⁽²⁸⁾

- (A) Yes
- (B) Yes, porosity and profile are acceptable
- ☒ (C) No, excessive porosity exceeding limit and excessive reinforcement
- (D) No, grinding is not permitted for weld repair
- (E) None of the above

(17)



The weld EG1-4 was made in cyclically loaded connection. Is this weld acceptable? ⁽²⁹⁾

- (A) Yes, only one pore
- (B) Yes, since there are no gouge or defects
- (C) Yes, since it is approved by forman
- ☒ (D) No, pore size exceeding 2.5mm (3/32 inch)
- (E) Non of the above

Edge EE1 was made for GMAW. Is the roughness of this edge acceptable? ⁽³⁰⁾

- (A) Yes, it is less than sample 3
- (B) Yes, roughness is acceptable
- ☒ (C) No, since it is gouged by 2 mm
- (D) Yes, it is sample 4 roughness
- (E) None of the above

(19)

Edge EE-2 was cut on a member not subjected to calculated stress. Is this acceptable? ⁽³¹⁾

- (A) Yes, roughness does not matter in this case
- ☒ (B) Yes it's roughness is sample 2
- (C) No, it rougher than sample 2
- (D) Yes it's roughness is sample 1
- (E) Non of the above

(20)

Edge EE-3 was prepared for SAW process. Is it acceptable? ⁽³²⁾

- (A) Yes, roughness and gouges are acceptable
- ☒ (B) No, it is gouged
- (C) Yes, roughness is sample 3
- (D) Yes, roughness is sample 2
- (E) None of the above

(21)

What is the effective length of weld wrapping around edge A if the specified size is 4.7mm (3/16 inch)? ⁽³³⁾

- (A) 95 mm
- (B) 90 mm
- ☒ (C) 80 mm
- (D) 70 mm
- (E) None of the above

(22)



What is the effective length of the weld wrapping corner B if specified size is 4.7 mm (3/16 inch)? ⁽³⁴⁾

- (A) 4.7 mm
- (B) 20 mm
- (C) 34 mm
- ☒ (D) 45 mm
- (E) 59 mm

Consider fillet weld undersize tolerances.

(23)



What is the effective length of fillet weld (EF-2) in the lap joint, if the specified size is 4.7 mm (3/16 inch)? ⁽²⁴⁾

- (A) 52 mm
- (B) 43 mm
- (C) 39 mm
- ☒ (D) 32 mm
- (E) Non of the above

(24)

New

Structural Steel

- Examine the 13 mm fillet weld of the T-joint EF-5.

Which one of the following is true regarding the visual inspection quality of this joint?

- (A) accepted
- **(B) unacceptable – convexity**
- (C) unacceptable – undersize
- (D) unacceptable – spatter
- (E) unacceptable – crater not filled

Leg = 12.7 mm

Actual throat = 15.9 mm

Theoretical throat = $12.7 / 1.41 \sim 9 \text{ mm}$

Convexity = $15.9 - 9 = \underline{6.9 \text{ mm}}$

Weld width = $12.7 * 1.41 \sim 18 \text{ mm}$.

Max convexity = 3 mm

Convexity > allowable - **unacceptable**

Size is 12.7 mm – accepted

(min size is 10 mm)

Spatter – accepted for VT

Throat at crater is 12 mm – accepted

(theoretical 9 mm)

(25)

New

Structural Steel

- Examine the 12 mm fillet weld of the T-joint EF-3.

Which one of the following is true regarding the visual inspection quality of this joint?

- (A) unacceptable – convexity
- (B) unacceptable – UC
- (C) unacceptable – crater not filled
- (D) unacceptable – crack
- **(E) unacceptable – convexity, crack**

Leg = 10.5 mm

Actual throat = 15.0 mm

Theoretical throat = $10.5 / 1.41 \sim 7.5$ mm)

Convexity = $15.0 - 7.5 = \underline{7.5}$ mm

Weld width = $10.5 * 1.41 \sim 15$ mm.

Max convexity = 3 mm

Convexity > allowable – unacceptable

UC < 0.5 mm – accepted

Throat at crater is 12.5 mm – accepted
(theoretical 7.5 mm)

Crater crack – unacceptable

(26)

New

Structural Steel

- Examine the 16 mm fillet weld of the T-joint EF-2.

Which one of the following is true regarding the quality of this joint?

- (A) accepted
- (B) unaccepted – undersize
- (C) unaccepted – crack
- (D) unaccepted – oversize
- (E) none of the above

Leg = 15.0 mm

Actual throat = 7.0 mm

Theoretical throat = $15.0/1.41 \sim 10.6$ mm)

Actual throat < theoretical (undersize)

(27)

New

Structural Steel

■ Examine the 10 mm fillet weld of the T-joint EF-4.
Which one is true regarding the quality of this joint?

- (A) unaccepted – convexity
- (B) unaccepted – undersize
- (C) unaccepted – arc strike
- **(D)** all of the above
- (E) accepted

Leg at left side ~ 8 mm
Actual throat at left side = 11.0 mm
Theoretical throat = $8.0/1.41 \sim 5.7$ mm)
Convexity = $11.0 - 5.7 = \underline{5.3}$ mm
Weld width = $8.0 * 1.41 \sim 11.5$ mm.
Max convexity = 3 mm
Convexity > allowable – unaccepted

Leg at right = 5 mm = undersize (min = 7 mm)
= unaccepted
Arc strike – unaccepted

(28)

New

Structural Steel

▪ Examine the 15 mm fillet weld of the T-joint EF-1.
Which one is true regarding the quality of this joint?

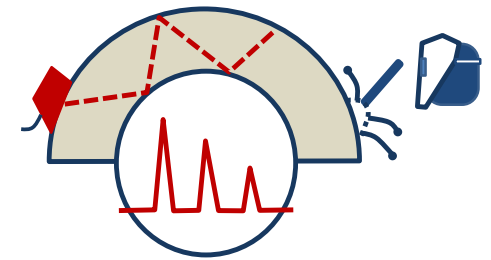
- (A) unacceptable – undersize
- (B) unacceptable – overlap
- (C) unacceptable – arc strike
- (D) all of the above
- (E) only A and B

Min size = 12 mm

Leg (size) ~ 6 mm – undersize unacceptable

Overlap at lower plate – unacceptable

PIPELINE CRITERIA



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(29)

For fitted pipes specimen (P2-3), what is the internal high-low at segment (4)? Is this acceptable as pipeline? ⁽⁷⁾

- ☒ (A) 2 mm, acceptable
- (B) 3 mm, acceptable
- (C) 4 mm, acceptable
- (D) 5 mm, acceptable
- (E) 6 mm, unacceptable

(30)



For fitted pipes specimen (P2-3), what is the internal high-low at segment (5)? Is this acceptable as pipeline? ⁽⁸⁾

- ☒ (A) 1 mm , acceptable
- (B) 1.5 mm, acceptable
- (C) 3 mm, acceptable
- (D) 4 mm, acceptable
- (E) none of the above

(31)

Examine groove EG2-5 of pipeline criteria. Which one is true? ⁽³⁴⁾

- ☒ (A) rejected – unrepaired shallow crack
- ☐ (B) rejected – cluster porosity
- ☐ (C) rejected – arc burn
- ☐ (D) rejected – both B and C
- ☐ (E) accepted

(32)



Examine groove EG2-4 of pipeline criteria. Which one is true, if the fitted groove width is 25 mm? ⁽³⁵⁾

- (A) rejected – lack of fusion
- (B) rejected – under cut (UC)
- (C) rejected – crown height is smaller than the specified.
- ☒ (D) rejected – the weld face is wider than the specified
- (E) both B and D

(33)

Examine midweld P2-2 between marks 4 and 6 of pipeline. Weld quality is? ⁽³⁶⁾

- (A) accepted
- (B) rejected – EU depth
- (C) rejected – CP diameter
- (D) rejected – IF length
- (E) both B and C

Unacceptable if IF > 25 mm, or
IF > 22 mm for pipe diameter of 88 mm

(34)

Examine upstream of pipeline at markers 1-5, based on thickness 8.7 mm. The weld quality is? ⁽³⁷⁾

- (A) accepted
- (B) rejected – EU depth
- (C) rejected – beads start and terminate at same location
- (D) rejected – spatter
- ☒ (E) both C and D

(35)

Examine downstream weld of pipe welded with SMAW using WPS SMAW-1 for pipeline, where thickness is 8.7 mm. The weld quality is? ⁽³⁸⁾

- (A) rejected – EU depth
- (B) rejected – excessive crown
- (C) rejected – spatter
- (D) all above
- ☒ (E) only B and C

(36)

According to acceptance criteria of pipeline, the specimen (P1-3), the weld at segment (9) is: ⁽⁹⁾

- (A) Acceptable because it is free from defects
- (B) Unacceptable because of arc burns, porosity and overlap
- (C) Unacceptable because of spattering and overlap
- ☒ (D) Unacceptable because of arc burns, excessive reinforcements, and adjacent beads started at same location (non-uniform cross-section)
- (E) None of the above

(37)



According to pipeline acceptance criteria for inadequate penetration.
The weld P2-1 is: ⁽¹²⁾

- (A) Unacceptable because IPD exceeds 50 mm
- (B) Acceptable because IPD does not exceeds 50 mm
- (C) Acceptable because the aggregate IPD does not exceed 75 mm
- (D) Acceptable because there is no IP
- ☒ (E) both of B and C

(38)

- Examine the EG1-3 weld for pipeline criteria. After excavation of defects, the groove area shall be:
 - (A) visually examined
 - (B) examined by PT or MT
 - (C) examined by RT
 - (D) A and B above
 - (E) A and C above

(39)

New

Pipeline

- Examine the downstream pipe at markers 0-6 for pipeline criteria. During welding, the inspector noted a shallow cracks at the incomplete weld. Which action is correct regarding this?
- (A) accepted
- (B) unaccepted
- (C) remove and reweld
- (D) ask for company authorization of the repair
- (E) all shallow cracks are acceptable

Rework is not a repair and does not require company authorization

(40)

New

Pipeline

- Examine the EG1-2 weld and porosity at the right-hand side of the joint for pipeline criteria. Which one is correct regarding this?
- (A) accepted CP less than 13 mm
- (B) unaccepted
- (C) remove and reweld
- (D) ask for company authorization of the repair
- (E) depth of porosity is unaccepted

(41)

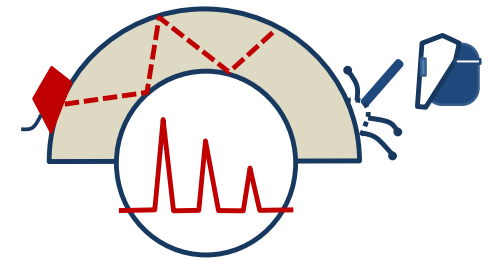
New

Pipeline

- Examine the EG1-2 weld and porosity at the left-hand side of the joint for pipeline criteria. Which one is correct regarding this for W.T. = 20 mm?
- (A) accepted
- (B) unaccepted
- (C) remove and reweld
- (D) ask for more NDE
- (E) depth of porosity is unaccepted

Pore size > 3 mm, and
CP diameter > 13 mm

PRESSURE PIPING CRITERIA



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(42)

Examine B2 for pressure piping. Which one is correct disposition of this B2? ⁽²³⁾

- (A) disregarded – incorrect grinding
- (B) disregarded – improper edge radius
- (C) disregarded – weld is not centered
- ☒ (D) reject – defect > 3 mm
- (E) reject – open defect > 6 mm

(43)

Examine the partially completed downstream weld of pressure piping at markers 4-5, where thickness is 8.7 mm. Is there sufficient weld deposit to allow interruption of the 95°C preheat and welding?

(³⁹)

- ☒ (A) yes
- (B) no – filling < 25%
- (C) no – filling < 50%
- (D) no – filling < 75%
- (E) no – groove must be completely filled

(44)

New

Pressure piping

■ Examine the downstream welded pipe at markers 6-11, for a single welded butt joint of 150 mm schedule 80 to 150 mm schedule 120 pipes. The design temperature is 450°C. Which one of the following is true for the reinforcement this joint?

- (A) accepted
- (B) unaccepted – > 2 mm
- (C) unaccepted – > 2.5 mm
- (D) unaccepted – > 3 mm
- (E) reinforcement shall be removed

- Weld reinforcement shall be based on the thickness of the thinner of materials joined.
- It shall be determined from the higher of the abutting surfaces involved.

(45)

New

Pressure piping

- Examine the midweld stainless steel pipe at markers 3-5, for a single welded butt joint of 200 mm schedule 10S to 200 mm schedule 40S. The design temperature is 470°C. Which one of the following is true for the quality of this joint?
- (A) accepted
- (B) unaccepted – reduction of at weld toe
- (C) unaccepted – reinforcement
- (D) unaccepted – linear indication > 5 mm
- (E) all the above

(46)

New

Pressure piping

- Examine bend B5 for pressure piping application, which one is true regarding B5?
- (A) accepted
- (B) disregarded – improper edge radius
- (C) disregarded – incorrect grinding
- (D) disregarded – incorrect width
- (E) reject – defect > 3 mm

(47)

Examine the upstream pipe of pressure piping, plus VT which tests of inspection shall be done for the finished surface after repair welding is completed of the excavation in this weld? ⁽⁹⁾

(A) no additional tests

(B) PT

(C) MT

(D) UT

☒ (E) B or C

(48)

Examine the down stream partially welded pipe between marker 2 and 3 of pressure piping, which one is correct? ⁽¹¹⁾

- (A) tack weld must be removed لو فيه شروخ
- (B) tack weld may be incorporated in weld
- (C) tack weld end shall be ground before proceeding لو مفيش شروخ
- (D) tack weld is irrelevant if the other tacks are sound
- ☒ (E) both B and C

(49)

Examine the middle weld on pressure piping between marker 5 and 10, where the thickness is 8.7 mm, design temp. 150°C, the weld quality is: ⁽⁸⁾

- (A) accepted
- ☒ (B) rejected – IF
- (C) Rejected – spatter
- (D) rejected – excess reinforcement
- (E) both B and C

IF is the cause of rejection, spatter is acceptable since it is not on the BM

(50)



Examine the internal alignment of partially welded pipe at marker 4 of pressure piping. Which one is true? ⁽¹²⁾

- (A) rejected - > 2 mm
- (B) rejected - > 0.8 mm and < 2 mm
- (C) rejected - > 2.5 mm and < 5 mm
- (D) accepted - > 2.5 mm and < 3 mm
- ☒ (E) accepted - < 2 mm

(51)

For pressure piping, welding may be interrupted provided that: ⁽¹⁾

- (A) Minimum 10 mm thickness of weld is deposited.
- (B) 25% of groove is filled.
- ☒ (C) A and B whichever is less.
- (D) Welding should not be interrupted until the joint is completed.
- (E) None of the above.

(52)

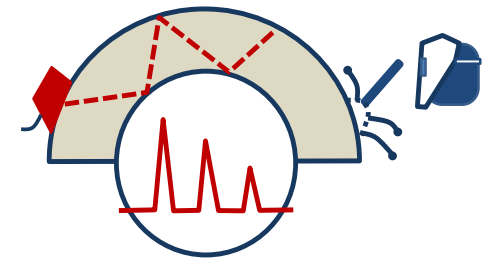


Pressure piping

For fitted pipes specimen (P2-3), what is the internal high-low at segment (3)? Is this acceptable as pressure piping? ⁽⁶⁾

- (A) 1 mm , acceptable
- ☒ (B) 1.5 mm, acceptable
- (C) 3 mm, acceptable
- (D) 4 mm, acceptable
- (E) none of the above

PROCEDURE QUALIFICATION WPS — PQR



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(53)

Examine exhibit PQR No. 002. Which WPS variable is not qualified by the PQR for: ⁽⁴²⁾

process: FCAW

material: A369 Gr FP1

pipe size: 200 mm

pipe thickness: XXS

weld: V-groove with backing

position: flat

filler: E491T (E71T)

preheat/interpass: 95°C/260°C

PWHT: N.A.

shielding gas: 75%Ar + 25%CO₂

- (A) not qualified – position and material
- (B) not qualified – material and filler metal
- (C) not qualified – gas composition and flow rate
- (D) not qualified – filler metal and preheat/interpass temperature
- ☒ (E) not qualified – material and preheat/interpass temperature

(54)

Examine exhibit WPS P91-4 and P91 heat treatment chart. Is the holding time comply with WPS for 100 mm thickness? ⁽⁴³⁾

- (A) Yes
- (B) No – holding time < WPS
- ☒ (C) No – holding time > WPS
- (D) No – WPS does not permit PWHT
- (E) No – it is < 15 minutes

(55)

Examine the upstream pipe edge preparation bevel angle for compliance with the the exhibit WPS GMAW -3. Which one is true regarding the edge preparation? ⁽²⁹⁾

- (A) reject – single bevel edge shape not permitted by the WPS
- (B) reject – bevel angle $>$ permitted by the WPS
- ☒ (C) reject – bevel angle $<$ permitted by the WPS
- (D) both B and C – bevel angle varies between markers 7 and 8
- (E) accepted bevel angle per WPS

(56)

Examine the root opening of the pipe downstream weld at marker 1-6 for compliance with exhibit 8 WPS 6. What is describing the root opening? ⁽³⁰⁾

- (A) reject – due to > 5 mm
- (B) reject – due to < 2.5 mm
- ☒ (C) reject – due to > 2.5
- (D) reject – due to > 2 mm and < 2.5 mm
- (E) accepted

(57)

Examine PWHT chart of exhibit # 6, what vlaue must be entered on the line 38 of PQR shown in Annex VIII? ⁽²⁰⁾

- (A) 1350 F
- ☒ (B) 1400 F
- (C) 1450 F
- (D) 3 hours
- (E) 5 hours

(58)



An edge preparation of P1-4 was performed according to WPS 6 (exhibit # 8). Is this is acceptable edge preparation? ⁽¹⁹⁾

- (A) Yes
- ☒ (B) No, because counter bore depth is less than 12.5mm
- (C) Yes, counterbore depth and thickness are acceptable.
- (D) No, the edge is not for WPS 6
- (E) None of the above

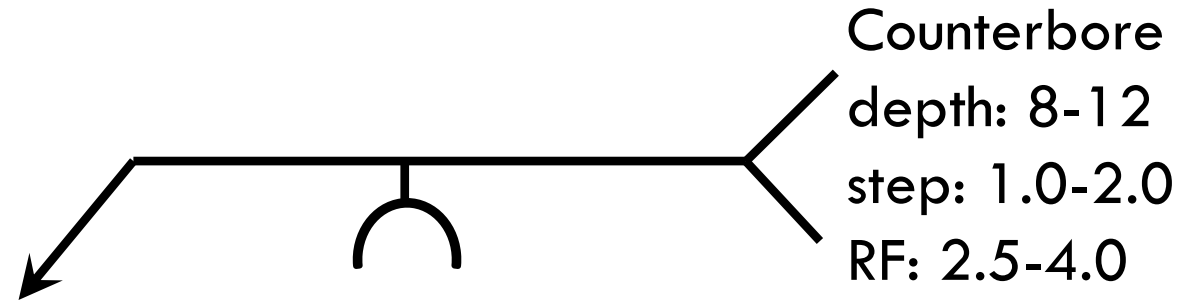
(59)

New

WPS – PQR

Examine the edge preparation of P1-4 and the the welding symbol. Is this is acceptable edge preparation? ⁽⁴²⁾

- (A) Yes
- (B) No – counterbore depth
- ☒ (C) No – counterbore step
- (D) No – root face
- (E) None of the above



(60)



Examine the tensile test calculations for the ultimate unit stress on PQR No. 007B. What are the correct ultimate unit stress values for the specimens 1 and 2? ⁽⁴¹⁾

- | | |
|--|------------------------|
| (A) specimen 1: 16,056 lbs | specimen 2: 14,939 lbs |
| (B) specimen 1: 103,042 psi | specimen 2: 94,279 psi |
| ☒ (C) specimen 1: 102,924 psi | specimen 2: 94,551 psi |
| (D) specimen 1: 103,238 psi | specimen 2: 94,077 psi |
| (E) specimen 1: 104,088 psi | specimen 2: 95,212 psi |

(61)

Weld specimen P2-3 was used for procedure qualification test (thickness = 7mm) for GTAW welding in the root pass and SMAW for filling and cap layers. What the range of base metal thickness qualified in each process? Measurements [1.5 mm GTAW- and 5.5 mm SMAW]. ⁽¹⁶⁾

- (A) 0.75 to 3 mm for GTAW and 5 to 11 mm for SMAW
- ☒ (B) 0.75 to 3 mm for GTAW and 2 to 11 mm for SMAW
- (C) 11 mm for both
- (D) 7 mm for GTAW and 11 mm for SMAW
- (E) None of the above

(62)

Weld specimen P2-3 was used for procedure qualification test (thickness = 7mm) for GTAW welding in root pass and SMAW filling and cap layers. What the range range of deposited weld metal thickness qualified in each process? Measurements [1.5 mm GTAW and 5.5 mm SMAW]. ⁽¹⁷⁾

- ☒ (A) Max of 3 mm for GTAW and 11 mm for SMAW
- (B) Max 1.5 mm for GTAW and 5.5 mm for SMAW
- (C) 0.75-3 mm for GTAW and 2-14 mm for SMAW
- (D) 0.75-3 mm for GTAW and 2-7 mm for SMAW
- (E) None of the above

(63)

The qualification of welding procedure shown in exhibit # 10 PQR No.003 was conducted for the WPS GMAW-3 (exhibit # 4). The P2-0 sample was used for PQR test. Is the edge parameter acceptable? ⁽¹⁰⁾

- (A) Yes, since bevel angle and root face are acceptable.
- (B) No, angle is 20° (less than the specified)
- (C) No, angle is acceptable and root face is 3 mm
- (D) No, root face is 3 mm (larger than the specified),
- ☒ (E) Both of B and D

(64)

Examine exhibit WPS FCAW-2. During an in process inspection, the inspector noted that:

- welding position: flat, horizontal, vertical
- interpass temperature : 120°C
- wire speed: 4.8 m/min
- shielding gas: 75% Ar + 25% CO₂ at 21 l/min
- electrode dia.: 1.2 mm
- electrode type: **E490T-1M** (E70T-1M)

Is the welder in compliance with this WPS? ⁽⁴⁰⁾

- ☒ (A) Yes
- (B) no – interpass temperature
- (C) no – wire speed
- (D) no – welding position
- (E) both B and D

(65)

In production welding of structural steel according to WPS SMAW-1 (exhibit # 1), the edge preparation of single bevel shown in specimen edges EG2-0 and EG2-6 are: ⁽¹¹⁾

- (A) The groove radius is acceptable
- (B) Unacceptable angle and root face
- (C) Acceptable angles and root face are conforming to the specified
- (D) Acceptable and are within range
- (E) Unacceptable, because of angles**

- Angles are 50 and 20, root faces are 0 and 0
- Root faces are acceptable.
- Angles are not.

(66)

A qualification test of 10 mm (3/8 inch) thick plate for welding procedure was done using GMAW-S process. The range of qualification is: ⁽²¹⁾

- ☒ (A) 11 mm
- (B) 22 mm
- (C) 50 mm
- (D) 80 mm
- (E) Non of the above

(67)

A qualification test of 16 mm thick plate for welding procedure was done using GMAW-S process. The range of qualification is: ⁽²²⁾

- (A) 2 – 17.6
- (B) 2 - 22
- ☒ (C) 5 - 32
- (D) 5 – 17.6
- (E) Non of the above

(68)

In Welder qualification test on 4 mm thick plate (groove) was done using GMAW-S process. The max. qualified deposit thickness is: ⁽²³⁾

- (A) 4.0 mm
- ☒ (B) 4.4 mm
- (C) 8.0 mm
- (D) unlimited
- (E) none of the above

(69)

In a production welding according to WPS GMAW-S (exhibit # 4) which is qualified in PQR No.003 (exhibit # 10)? The preheat temp. was 75°F. Is this acceptable? ⁽³⁵⁾

- (A) Yes, above 50°F
- ☒ (B) No, more than 100°F (55°C) below qualified preheat temperature.
- (C) No, it is 40°F above the WPS preheat temperature range
- (D) Yes, preheat is optional
- (E) None of the above

(70)

Which of the following is true regarding the bakeout treatment shown in exhibit # 6? ⁽³⁶⁾

- (A) The bakeout temperature is not attained
- (B) The holding is not suitable
- (C) Cooling to room temperature for NDE was not executed
- ☒ (D) Both A and C
- (E) None of the above

(71)

Are heating and cooling rates of PWHT of chart (exhibit # 6) are acceptable within the limits of WPS P91-4 (exhibit # 5)? ⁽³⁸⁾

- ☒ (A) Yes, both are lower than the max. specified
- ☐ (B) No, both are larger than the specified
- ☐ (C) Cooling rate is higher than specified
- ☐ (D) Heating rate is not critical and may be of any value.
- ☐ (E) Non of the above

(72)

Examine exhibit PQR No. 002. A WPS based on this PQR would be qualified for what base metal thickness range? ⁽¹⁴⁾

- (A) 5 mm – 50 mm
- (B) 13 mm – 50 mm
- (C) 25 mm – 33.25 mm
- ☒ (D) 5 mm – 200 mm
- (E) 3 mm – unlimited

(73)

Examine the tensile test results on PQR No. 001. Annex III provides the following:

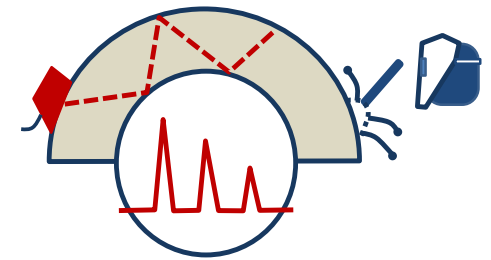
Material A106 Gr C

Min Tensile: 485 MPa

which of the following is true? ⁽¹⁶⁾

- (A) accepted
- ☒ (B) rejected – weld metal failure is below the base metal SMYS
- (C) rejected – both values below the base metal SMYS
- (D) rejected – one value more than 5% below metal SMYS
- (E) rejected – both values more than 5% below metal SMYS

PERFORMANCE QUALIFICATION WQTR



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(74)

Examine the exhibit WQTR – B, for which material the welder is qualified? ⁽²⁴⁾

- (A) A36
- (B) A106 Gr C
- (C) API 5L X80
- ☒ (D) All the above
- (E) Only A or B

(75)

New

Performance Qualification

- Examine exhibit WQTR-A. Revise the variables:
 - * process type: manual
 - * electrode spec.: A5.18
 - * vertical prog.: uphill
 - * Base metal qualified : M 1
 - * F-number: 4
- Which one is correct for these variables?
- (A) manual, M11, A5.18, 6, uphill
- (B) manual, M1-M11, A5.20, 6, downhill
- (C) semiautomatic, M1-M11, A5.20, 6, uphill
- (D) semiautomatic, M1, A5.20, 6, uphill/downhill
- (E) semiautomatic, M1-M11, A5.18, 6, uphill

F-NUMBER 6 مهم جدا

Spec	Process	Electrode	Material
AWS A5.9	GMAW, GTAW, PAW	ER308	Stainless steel
AWS A5.18	GMAW, GTAW, PAW	ER70S-6	Carbon steel
AWS A5.28	GMAW, GTAW, PAW	ER90S-B3	Low-alloy steel
AWS A5.20	FCAW	E71T-1C	Carbon steel
AWS A5.22	FCAW	E308LT	Stainless steel
AWS A5.29	FCAW	E91T1-B3L	Low-alloy steel
A5.30	GTAW	IN5XX	Consumable inserts

(76)

New

Performance Qualification

- Examine exhibit WQTR-G. Revise the variables:
 - * process type: manual
 - * base metal qualified : M 1-11 & 4X
 - * F-number: 6
 - * base metal M #: 8
 - * electrode spec.: A5.22
 - * Position qualified: *
- Which one is an acceptable correction for these variables?

- (A) manual, M1, M 1-11 & 4X, A5.22, 6, All
- (B) manual, M8, M1-11, A5.20, 6, F&H
- (C) semiautomatic, M1, M1-11, A5.22, 6, All
- (D) semiautomatic, M1, M 1-11, A5.20, 6, All
- (E) semiautomatic, M8, M1-M11 & M4X, A5.18, 6, F&H

M4X= nickel alloys
=M41, M42, M43, M44 etc.

(77)

New

Performance Qualification

- Examine exhibit WQTR-F. Revise the variables:
 - * process type: semiautomatic
 - * base metal M #: 8
 - * base metal qualified : M 1-11 & 4X
 - * electrode spec.: A5.22
 - * F-number: 6
 - * Position qualified: *
- Which one is an acceptable correction for these variables?

- (A) manual, M1, M 1-11 & 4X, A5.22, 6, All
- (B) manual, M8, M1-11, A5.20, 6, H
- (C) semiautomatic, M1, M1-11, A5.20, 6, All
- (D) semiautomatic, M8, M 1-11, A5.20, 6, All
- (E) semiautomatic, M8, M1-M11 & M4X, A5.18, 6, F & H

M4X = nickel alloys
= M41, M42, M43, M44 etc.

(78)

New

Performance Qualification

- Examine exhibit WQTR-E. Revise the variables:
 - * process type: manual
 - * base metal qualified : M 1-11 & 4X
 - * Electrode classification: E71T-1C, E71T-1m
 - * base metal M #: 8
 - * electrode spec.: A5.22
 - * Position qualified: *
- Which one is an acceptable correction for these variables?

- (A) manual, M1, M 1-11 & 4X, A5.22, E71T-1M, All
- (B) manual, M8, M1-11, A5.20, E71T-1M, F
- (C) semiautomatic, M1, M1-11, A5.20, E308LT, All
- (D) semiautomatic, M8, M 1-11, A5.22, E71T-1C, All
- (E) semiautomatic, M8, M1-M11, A5.22, E308LT, All

E71T-1C = E491T-1C

(79)

New

Performance Qualification

- Examine exhibit WQTR-D. Revise the variables:
 - * process type: manual
 - * base metal qualified : M 1-11 & 4X
 - * Electrode classification: ER308
 - * base metal M #: 8
 - * electrode spec.: A5.9
 - * Position qualified: *
- Which one is an acceptable correction for these variables?
- (A) manual, M1, M 1-11 & 4X, A5.22, E71T-1M, All
- (B) manual, M8, M1-11, A5.20, E71T-1M, F
- (C) semiautomatic, M8, M1-11, A5.9, ER308, All
- (D) semiautomatic, M8, M 1-11, A5.18, E71T-1C, All
- (E) semiautomatic, M8, M1-M11, A5.18, E308LT, F&V

(80)

Examine the Exhibit WQRT-B. Is the welder qualified to weld the following: ⁽²⁵⁾

Material: A36

Progression: Vertical uphill

Backing: metal backing

- (A) Yes
- (B) No, for A36
- ☒ (C) No, for vertical up
- (D) No, for backing
- (E) Both B & C

(81)

Examine the Exhibit WQRT-C. Is the welder qualified to weld the following: ⁽²⁶⁾

Electrode: E4310 (E6010)

Progression: Downhill

Deposite: 5 mm thickness

- (A) Yes
- (B) No, for E4310
- (C) No, for downhill
- (D) No, for deposit thickness
- ☒ (E) Both B & C

(82)

Examine the Exhibit WQRT-E and the pipe specimens. Which weld is this welder qualified using only the SMAW, W.T. = 7 mm? ⁽²⁷⁾

- (A) upstream weld
- (B) middle weld
- (C) downstream weld
- ☒ (D) none above
- (E) only A and C

(83)

Examine the Exhibit WQRT-A. Is the welder qualified to weld the following: ⁽³⁹⁾

Material: A516 Gr 65

Weld: fillet

Position: horizontal

- ☒ (A) Yes
- (B) No, for A516 Gr 65
- (C) No, for fillet weld
- (D) No, for horizontal position
- (E) Both C & D

(84)

Examine the Exhibit WQRT-B. It qualifies the welder for: ⁽⁴⁰⁾

- (A) GTAW
- (B) SMAW
- (C) FCAW
- ☒ (D) GMAW
- (E) All of the above

(85)

Examine exhibit PQR No. 001 for steel base material.

For a welder qualification record WQTR based on this PQR:

The welder is not qualified to weld which one? ⁽²²⁾

(A) M1 Gr 2

(B) M11B

☒ (C) M21

(D) M4

(E) M5C

(86)

New

Performance Qualification

▪ Examine the midweld stainless steel pipe for performance qualification at markers 3-5, for a single welded butt joint of 200 mm schedule 10S (lower pipe) to 200 mm schedule 40S (upper pipe). Which one of the following is true for the quality of this joint?

- (A) accepted
- (B) unacceptable – UC
- (C) unacceptable – reinforcement
- (D) unacceptable – linear indication > 5 mm
- (E) all the above

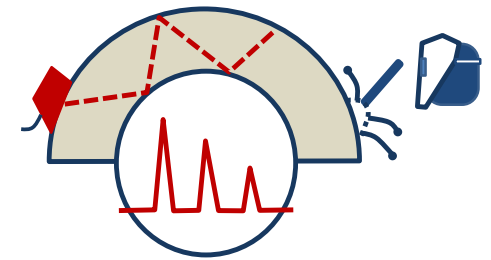
- Pipe thicknesses = 3.76 mm lower and 8.18 mm upper
- UC > 0.376 of lower pipe – **unaccepted**

(87)

Weld specimen P2-3 was used for welder qualification test (thickness = 7mm) for welder 1 using GTAW in root pass and welder 2 using SMAW in filling and cap layers. What the range of deposit weld metal thickness qualified for each welder? Measurements [1.5 mm GTAW-welder 1 and 5.5 mm SMAW-welder 2]. ⁽¹⁸⁾

- (A) 6 mm for welder 1, and 22 mm for welder 2
- (B) 3 mm for welder 1, and 5.5 mm for welder 2
- ☒ (C) 3 mm for welder 1, and 11 mm for welder 2
- (D) 1.5 mm for welder 1, and 11 mm for welder 2
- (E) None of the above

NDT TECHNIQUES

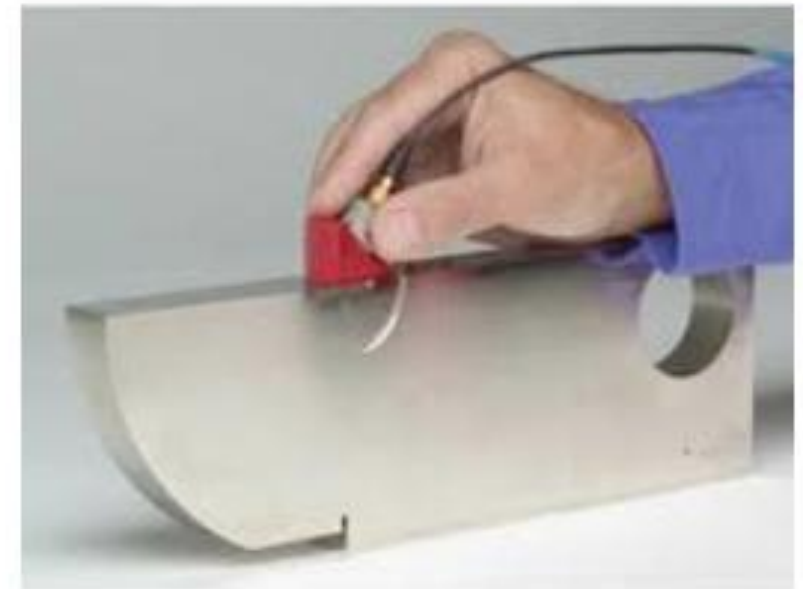


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(88)

Figure 4 shows which of the following procedures: ⁽⁴¹⁾

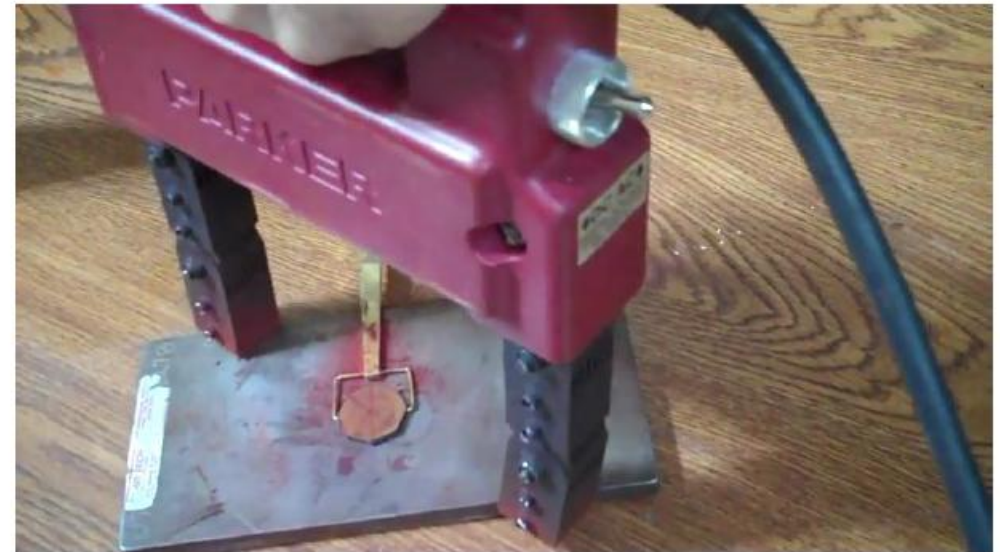
- (A) Ultrasonic calibration using V1 or V2 block
- (B) Magnetic particle testing calibration.
- (C) Angle beam calibration.
- ☒ (D) Both A and C.
- (E) None of the above.



(89)

Figure 1 shows which of the following procedures: ⁽²⁾

- (A) Ultrasonic calibration using V1, V2 blocks
- ☒ (B) Magnetic particle testing calibration.
- (C) Angle beam calibration.
- (D) Both A and C.
- (E) None of the above.



(90)

Figure 2 shows: ⁽³⁾

- (A) Longitudinal wave UT.
- ☒ (B) Shear wave UT.
- (C) Surface waves UT.
- (D) Both A and B
- (E) None of the above.



(91)

Examine Figure 1, the plastic bottle contains: ⁽²¹⁾

- (A) cleaning solvent
- (B) rust preventative
- (C) PT
- (D) high viscosity lubricant
- ☒ (E) UT couplant



(92)

Examine Exhibit Figure 2. What is performing: ⁽³¹⁾

- (A) scrubbing contamination off the transducer
- (B) angle beam calibration check
- ☒ (C) straight beam calibration check
- (D) getting the sensitivity level
- (E) demagnetizing the transducer

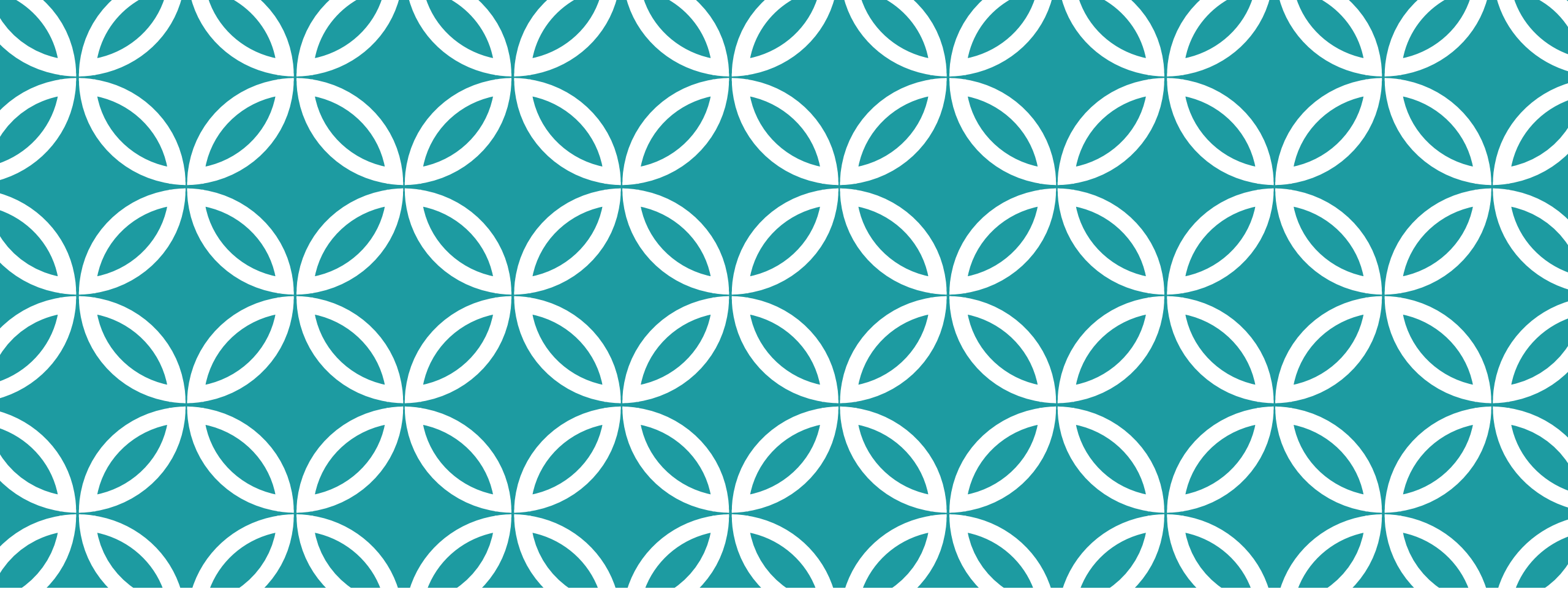


(93)

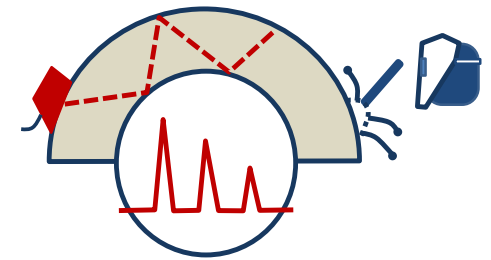
Figure 3 shows: ⁽⁴⁾

- (A) Longitudinal crack.
- (B) Crater crack.
- (C) Centerline crack.
- (D) Transverse crack.
- (E) None of the above.





ANNEXES AND OTHERS



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- Examine exhibit WPS P91-4 and Annex I of the BOS, the weld metal deposit chemistry of the E9015-B91 H4 electrode is:
 - C: 0.15% Cr: 9.30% Mo: 0.33%
 - Ni: 0.06% Mn: 0.88% Si: 1.08%
- Which one is true regarding the weld metal deposit chemistry?
 - (A) accepted
 - (B) unaccepted for A-5
 - (C) unaccepted for Cr level
 - (D) unaccepted for C level
 - (E) C and D above

(95)



Examine exhibit PQR No. 001 and Annex I of the BOS, the weld metal deposit chemistry of the E4918 (E7018) electrode is:

C: 0.04%	Cr: 0.05%	Mo: 0.0%
Ni: 0.06%	Mn: 0.60%	Si: 0.41%

which one is true regarding A number recorded in the PQR? ⁽¹⁷⁾

- (A) it is correct
- (B) incorrect should be 5
- (C) incorrect should be 4
- (D) incorrect should be 3
- (E) incorrect should be 2

(96)

Examine the partially completed P2-3 weld on the pipe between marker 12 and marker 1 on an 8.7 mm thickness. What is the approximate weld deposit thickness: ⁽¹⁾

- (A) 18 mm
- (B) 8.7 mm
- ☒ (C) 6.2 mm
- (D) 5.5 mm
- (E) 3.5

(97)

Examine the P1-0 end of the pipe. What is the pipe schedule at that end: ⁽⁴⁾

(A) 40

☒ (B) 80

(C) 120

(D) 160

(E) STD

(98)



A stainless steel pipe having an outside diameter of 114.3 mm and a wall thickness of 5.3 mm. What schedule is this pipe? ⁽⁵⁾

- (A) 10 s
- ☒ (B) 40 s
- (C) 80 s
- (D) 120 s
- (E) Non of the above

(99)

Examine bend B2 preparation: ⁽⁵⁾

- (A) accepted
- ☒ (B) unaccepted – due to width
- (C) unaccepted – edge radius
- (D) unaccepted – grinding direction
- (E) unaccepted – reinforcement

(100)

Examine the bend specimens. Which one is not ground as recommended? ⁽¹⁸⁾

- ☒ (A) B2
- ☐ (B) B3
- ☐ (C) B4
- ☐ (D) B5
- ☐ (E) B6

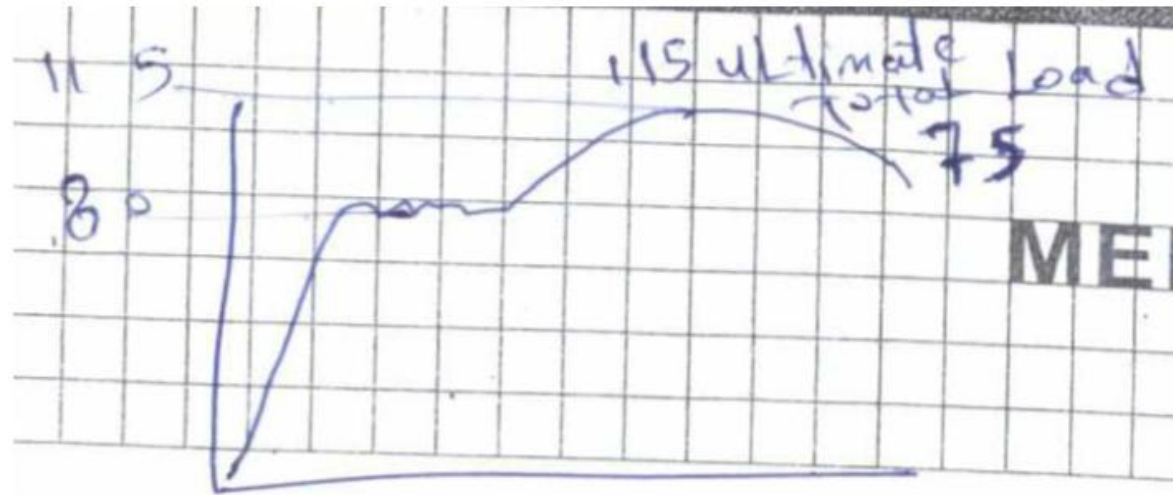
(101)

Examine bend B1, disregard any discontinuities, which one is true regarding preparation and bending? ⁽¹⁹⁾

- (A) accepted
- (B) rejected – weld not within bend portion
- ☒ (C) rejected – edge radius
- (D) rejected – grinding direction
- (E) both B and D

(102)

During a tensile test, it hesitated at 80 KN, then rise to a max load of 115 KN, then load dropped to 75 KN, then failed. What information must be recorded in space 57 of Annex VIII? ⁽¹³⁾



- (A) 75 KN
- (B) 80 KN
- ☒ (C) 115 KN
- (D) 155 KN
- (E) 275 KN

(103)

The initial specimen dimension of a rectangular tensile bar were 20 mm by 13 mm. What information should be recorded in space 56 of annex VIII? ⁽¹⁵⁾

- (A) 26
- (B) 33
- (C) 52
- ☒ (D) 260
- (E) 520

(104)

Weld # P1-1 was made in fixed position (5G) within 0'clock at top of pipe. The welding direction was: ⁽¹⁵⁾

- ☒ (A) Uphill
- ☐ (B) Downhill
- ☐ (C) Not clear
- ☐ (D) Both a and b

(105)



Weld # P2-1 was made in fixed position (5G) with 0'clock at top of pipe. The welding direction was: ⁽¹⁴⁾

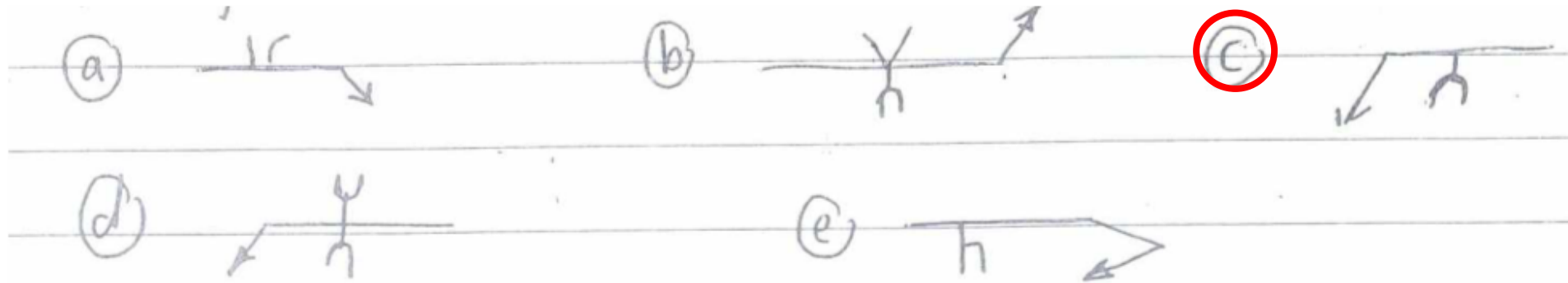
- (A) Uphill
- ☒ (B) Downhill
- (C) Not clear
- (D) Both a and b

(106)

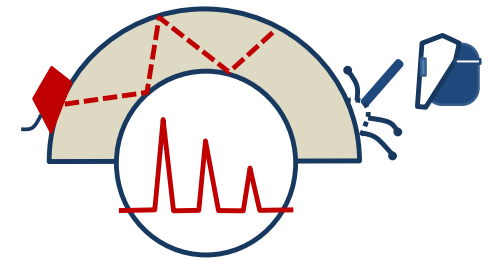
The welding directions in specimen EG2-1 are: ⁽¹³⁾

- (A) Left to right
- (B) right to left
- (C) right to left then left to right
- ☒ (D) Left to right then right to left
- (E) It is not possible to find weld directions from the specimen

Examine the downstream pipe edge preparation, assume a pipe with an identical edge preparation is fitted to this edge. Which of the following is correct? ⁽²⁸⁾



بالتوفيق



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